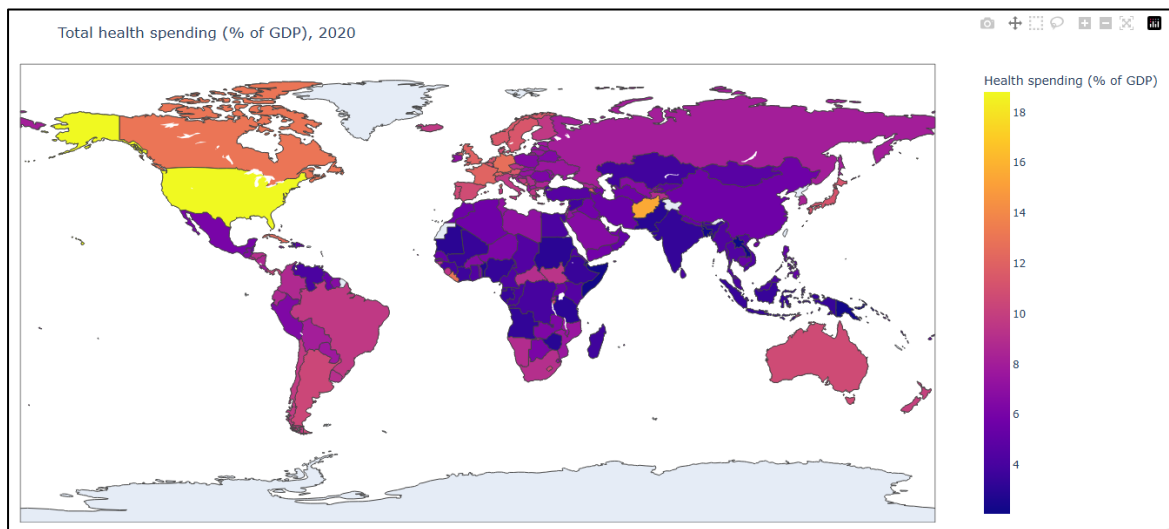


Title: Health Financing and Health Protection

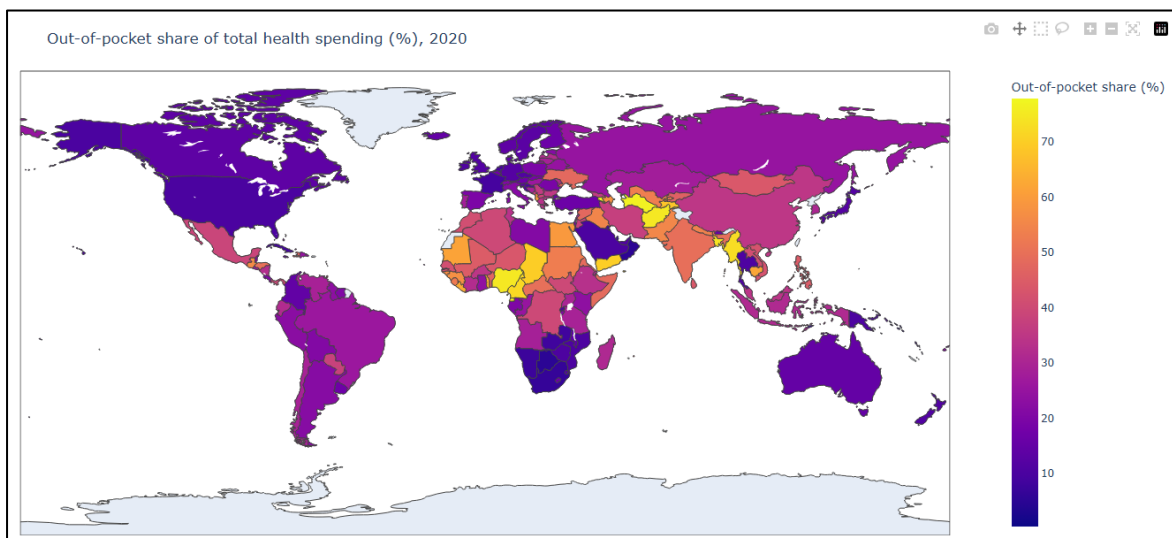
Name: Amerita Ravuvu

Date: 18/05/2026

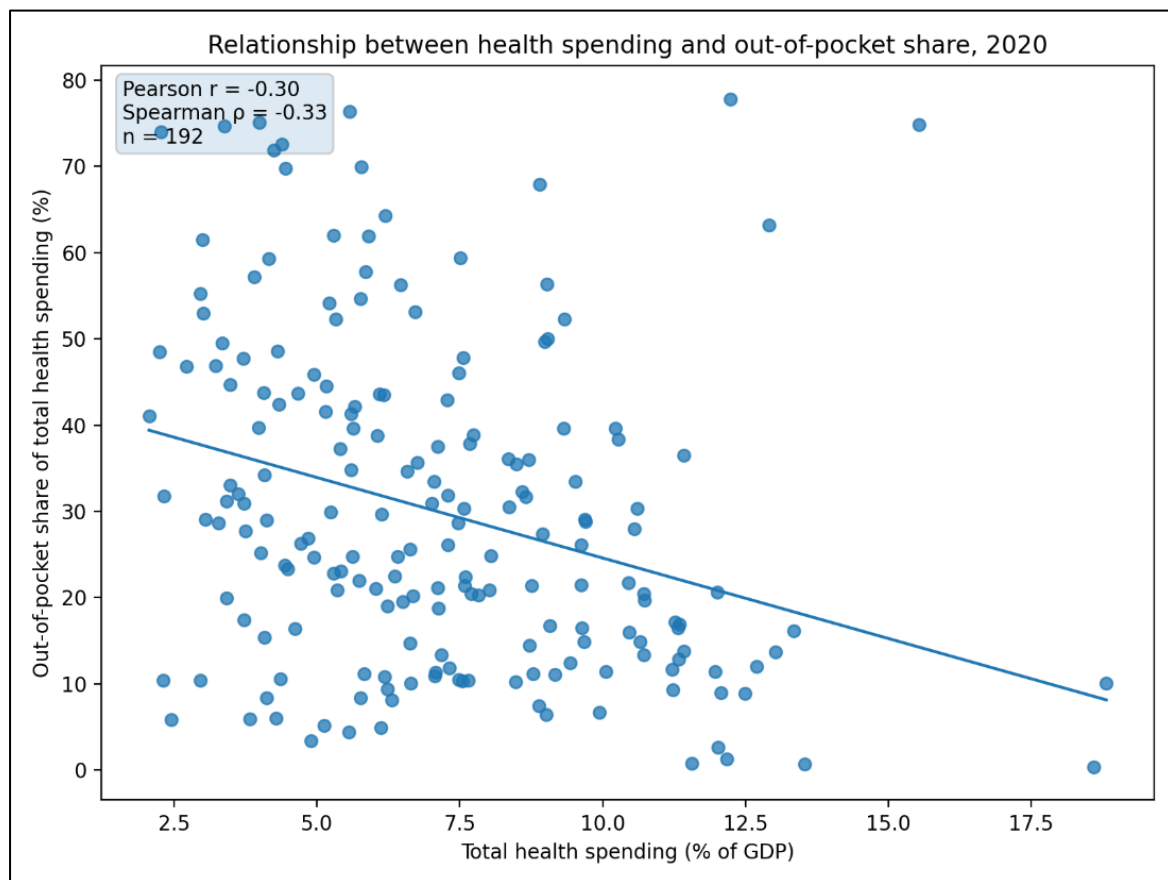
Total Health Spending (% of GDP), 2020



Out-of-pocket Share of Total Health Spending (%), 2020



Relationship Between Health Spending and Out-of-pocket Share, 2020



Main finding:

The 2020 relationship is moderately negative:

- Pearson correlation: -0.304
- Spearman correlation: -0.325
- Linear slope: about -1.87 percentage points in out-of-pocket share for each 1 percentage-point increase in total health spending as a share of GDP.

Interpretation:

Countries that spend a larger share of GDP on health tend, on average, to have a lower out-of-pocket share of total health spending. This suggests that higher overall health expenditure may be associated with stronger pooled or public financing, reducing the direct burden on households. However, the relationship is not very strong: there is substantial variation, meaning health financing structure, insurance coverage, public-sector capacity, and policy design likely matter in addition to total spending levels.

Reflection:

Working through this analysis highlighted a mix of straightforward and not so straightforward steps for me. Loading the datasets, checking their structure and extracting the 2020 values were smooth. But on my first try I was not able to generate the world maps with any colors, however, this had to do with my current chat gpt plan.

Regarding the datasets, both CSVs were already in a clean, wide format with consistent country codes and complete 2020 data which no doubt removed a layer of preprocessing that with manual inputs, would usually slow down cross-country comparisons.

It was interesting to see how chatgpt ensured that the two indicators aligned conceptually and structurally. Even though datasets looked tidy, watching chatgpts processes of merging them requires careful checking for hidden inconsistencies and the work that goes behind this in checking the duplication of country entries, mismatched codes, missing values etc.

The biggest takeaway for me is the gentle reminder that clean data structures dramatically speed up the analysis and a fuller appreciation of how LLMs such as chatgpt are really taking over the 'tedious' validation steps which are essential for trustworthy results.